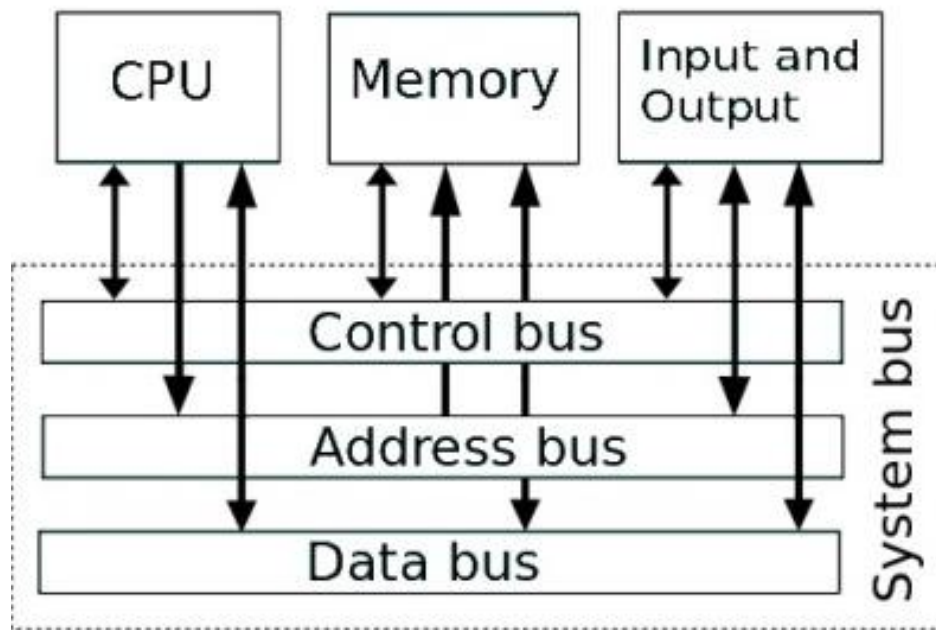


Lecture-3

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- Bit: A bit is a single binary digit.
- Word: A word refers to the basic data size or bit size that can be processed by the arithmetic and logic unit of the processor. A 16-bit binary number is called a word in a 16-bit processor.
- Bus: A bus is a group of wires/lines that carry similar information.
- System Bus: The system bus is a group of wires/lines used for communication between the microprocessor and peripherals.

- **Memory Word:** The number of bits that can be stored in a register or memory element is called a memory word.
 - **Address Bus:** It carries the address, which is a unique binary pattern used to identify a memory location or an I/O port. For example, an eight-bit address bus has eight lines and thus it can address $2^8 = 256$ different locations. The locations in hexadecimal format can be written as 00H – FFH.
 - **Data Bus:** The data bus is used to transfer data between memory and processor or between I/O device and processor. For example, an 8-bit processor will generally have an 8-bit data bus and a 16-bit processor will have 16-bit data bus.
 - **Control Bus:** The control bus carry control signals, which consists of signals for selection of memory or I/O device from the given address, direction of data transfer and synchronization of data transfer in case of slow devices.
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- **Functions of Control Unit of 8085:**

Generates signals on data bus, address bus and control bus within microprocessor to carry out the instruction, which has been decoded. Typical buses and their timing are described as follows:

- **Data Bus:** Data bus carries data in binary form between microprocessor and other external units such as memory. Data bus is **bidirectional** in nature. The data bus width of 8085 microprocessor is 8-bit i.e. 2^8 combination of binary digits and are typically identified as D0 – D7. Thus, size of the data bus determines what arithmetic can be done. If only 8-bit wide then largest number is 11111111 (255 in decimal).



- **Address Bus:** The address bus carries addresses and is one-way bus from microprocessor to the memory or other devices. 8085 microprocessors contain 16-bit address bus and are generally identified as A0 - A15. The higher order address lines (A8 – A15) are unidirectional and the lower order lines (A0 – A7) are multiplexed (time-shared) with the eight data bits (D0 – D7) and hence, they are bidirectional.

- **Control Bus:** The control bus carries control signals partly unidirectional and partly bidirectional. The following control and status signals are used by 8085 processor:

- I. ALE (Address Latch Enable):

$$\text{ALE} = \begin{cases} 1 \text{ (Eable)} & \text{AD}_0\text{-AD}_7 = \text{A}_0 - \text{A}_7 \text{ (Will carry only address).} \\ 0 \text{ (disable)} & \text{AD}_0\text{-AD}_7 = \text{D}_0 - \text{D}_7 \text{ (Will carry only data).} \end{cases}$$

- II. $\overline{RD}$ (active low output): when the $\overline{RD} = 0$, it will do read operation.
- III. $\overline{WR}$ (active low output): when the $\overline{WR} = 0$, it will do write operation.
- IV. $IO/\overline{M}$: It is a signal that distinguishes between a memory operation and an I/O operation. When $IO/\overline{M} = 0$, it is a memory operation, and when $IO/\overline{M} = 1$, it is an I/O operation.
- V. $S1$ and $S0$: These are status signals used to specify the type of operation being performed; they are listed in Table 1.

Table 1 Status signals and associated operations.

S1	S0	States
0	0	Halt
0	1	Write
1	0	Read
1	1	Fetch